Columbia University
IEOR4703 – Monte Carlo Simulation Methods (Hirsa)
Term Project 2
Volatility-Volume-based Order Management Utilizing Statistical and Rule-based Techniques

Term Project 2 – Volatility-Volume-based Order Management

1 Introduction

`Buy low, sell high` is arguably the most well-known adage among traders for achieving profits in the market. It embodies an ideal strategy that, while straightforward in theory, proves challenging in practice. Recognizing this, once armed with a decision-making tool and a compelling signal, one might question why not to purchase at an even lower price for a `Buy` signal, or sell at a higher one for a `Sell` signal.

The objective of this study is to segment the orders in such a manner that ensures a high likelihood of their execution while minimizing slippage as much as possible. The strategy focuses on adjusting the slicing of orders and limit orders based on volatility and volume.

This adjustment will be achieved through the application of statistical methods or rule-based techniques that involve the mining and classification of historical time series data. The efficacy of any proposed method will be assessed by back-testing the strategy/methodology.

2 Data

Traders and quants utilize various types of market data, often referred to as Level I and Level II market data. Level II data offers more detailed information than Level I. Also known as market depth, Level II data reveals the quantity of shares, contracts, or lots available at each specific price level. Furthermore, it is frequently called the order book because it displays a range of orders that are pending and awaiting execution. Orders are executed when market participants agree to buy or sell at a specific price. The well-known `TAQ` dataset includes both trades and quotes, featuring bid and ask prices along with their respective volumes.

For this study/project, we will be working with 1-min open-high-low-close (`OHLC`) futures data with volume. The below data is a sample data for S&P 500 Mini futures (`ESH21`) on February 5, 2021. As shown, the first entity is date, hour, min, second followed by open, high, low, close of that minute and traded volume in that interval.

In this study/project, we will be analyzing 1-minute open-high-low-close (`OHLC`) futures data that includes volume. Presented below is an example dataset for S&P 500 Mini futures (`ESH21`) from February 5, 2021. As illustrated, the initial data point represents the date, hour, minute, and

second, followed by the open, high, low, and close prices for that minute, along with the trading volume within that time frame.

date/time	*open*	*high*	*low*	*close*	*volume*
2021.02.05.09 : 30 : 00,	3885.25,	3886.25,	3884.00,	3885.00,	9295
2021.02.05.09 : 31 : 00,	3885.00,	3885.25,	3882.00,	3882.00,	8117
2021.02.05.09 : 32 : 00,	3882.00,	3885.00,	3882.00,	3883.25,	7288
2021.02.05.09 : 33 : 00,	3883.25,	3884.50,	3881.50,	3881.50,	6187
2021.02.05.09 : 34 : 00,	3881.50,	3884.00,	3880.50,	3880.75,	8865
2021.02.05.09 : 35 : 00,	3880.75,	3881.75,	3879.50,	3881.00,	6854
2021.02.05.09 : 36 : 00,	3880.75,	3881.00,	3879.00,	3879.00,	6008
2021.02.05.09 : 37 : 00,	3879.00,	3880.00,	3877.25,	3879.75,	7425
2021.02.05.09 : 38 : 00,	3879.75,	3880.25,	3878.50,	3879.75,	3955
2021.02.05.09 : 39 : 00,	3880.00,	3880.75,	3879.00,	3879.50,	4309
2021.02.05.09 : 40 : 00,	3879.50,	3881.25,	3879.25,	3880.25,	4001
2021.02.05.09 : 41 : 00,	3880.25,	3881.75,	3880.00,	3881.00,	2943
2021.02.05.09 : 42 : 00,	3881.25,	3881.75,	3880.25,	3881.25,	2902
2021.02.05.09 : 43 : 00,	3881.25,	3881.50,	3879.75,	3880.00,	2474
2021.02.05.09 : 44 : 00,	3880.00,	3880.25,	3879.25,	3879.25,	3632
2021.02.05.09 : 45 : 00,	3879.50,	3881.50,	3879.25,	3881.00,	5210
2021.02.05.09 : 46 : 00,	3881.00,	3882.25,	3880.00,	3880.75,	6082
2021.02.05.09 : 47 : 00,	3881.00,	3882.25,	3880.75,	3882.00,	3094
2021.02.05.09 : 48 : 00,	3882.00,	3883.00,	3881.75,	3882.25,	3112
2021.02.05.09 : 49 : 00,	3882.00,	3883.25,	3881.25,	3881.50,	2847
2021.02.05.09 : 50 : 00,	3881.75,	3882.25,	3881.25,	3882.00,	2858
2021.02.05.09 : 51 : 00,	3882.25,	3882.75,	3881.50,	3881.50,	1933
2021.02.05.09 : 52 : 00,	3881.50,	3881.75,	3880.75,	3880.75,	3436
2021.02.05.09 : 53 : 00,	3881.00,	3881.50,	3879.75,	3880.25,	3387
2021.02.05.09 : 54 : 00,	3880.25,	3880.50,	3879.25,	3879.50,	4496
2021.02.05.09 : 55 : 00,	3879.50,	3879.75,	3875.50,	3875.75,	8458
2021.02.05.09 : 56 : 00,	3875.75,	3876.00,	3874.00,	3874.75,	8499
2021.02.05.09 : 57 : 00,	3874.50,	3876.00,	3874.50,	3875.25,	4059
2021.02.05.09 : 58 : 00,	3875.00,	3875.25,	3874.00,	3874.50,	2824
2021.02.05.09 : 59 : 00,	3874.50,	3875.50,	3874.50,	3875.00,	2595

If for any reason you wish to write it on a 5-min basis, you would pick the open of the first minute as the open for the 5-min interval, the close of the last minute as the close for the 5-min interval and maximum of the 5-min interval as high and minimum of the 5-min interval as low and write it as follows:

date/time	*open*	*high*	*low*	*close*	*volume*
2021.02.05.09 : 30 : 00,	3885.25,	3886.25,	3880.50,	3880.75,	39752
2021.02.05.09 : 35 : 00,	3880.75,	3881.75,	3877.25,	3879.50,	28551
2021.02.05.09 : 40 : 00,	3879.50,	3881.75,	3879.25,	3879.25,	15952
2021.02.05.09 : 45 : 00,	3879.50,	3883.25,	3879.25,	3881.50,	20345
2021.02.05.09 : 50 : 00,	3881.75,	3882.75,	3879.25,	3879.50,	16110
2021.02.05.09 : 55 : 00,	3879.50,	3879.75,	3874.00,	3875.00,	26435

Term Project 2 – Volatility/Volume-Based Order Management

2.1 Data

For this project, you are provided futures data for 10 different markets,

- Nasdaq (`NQ`) – Nasdaq Futures
- Mini (`ES`) – S&P 500 E-Mini Futures
- UST (`TY`) – 10-yr US Government T-note
- Bund (`RX`) – German 10-yr Bund Futures
- Copper (`HG`) – Copper Futures
- Gold (`GC`) – Gold Futures
- Silver (`SI`) – Silver Futures
- Soybeans (`S`) – Soybeans Futures
- CrudeOil (`CL`) – Crude Oil WTI Futures
- EURUSD (`EC`) – EURUSD Futures

You should just use the data with high volume and high number of trades. The first few weeks or months may not be relevant at all. This can easily be done by eye-balling and/or simply based on number of trades. Days with many missing data/trades should be discarded (for example the number of trading minutes should be more than 90% to be considered). Also your module should be generic enough that you can easily apply it to any other futures market.

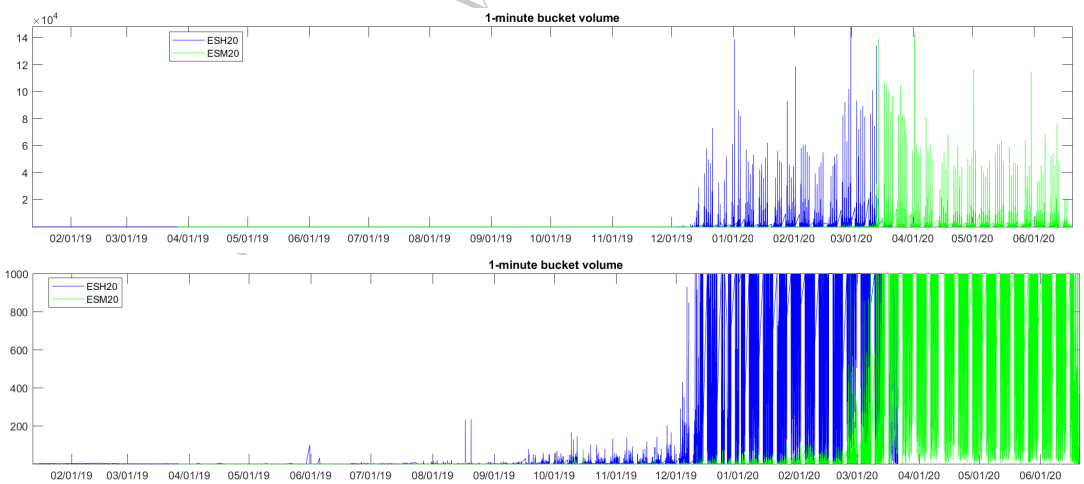

Figure 1: top plot: 1-min bucket volume for S&P 500 Mini futures contract for March 2020 (`ESH20`) and June 2020 (`ESM20`) expiry; bottom plot: zoomed-in version of the plot on the top

In Figure 1, we illustrate one-minute bucket volume for S&P 500 Mini futures contract for March 2020 (`ESH20`) and June 2020 (`ESM20`) expiry. The plot in the bottom of the figure is the zoomed-in version of it. As you can see before `ESH20` expires `ESM20` becomes more liquid around which takes place around March 17, 2020. For this data one would use `ESH20` from around 12/15/2019 to around 3/17/2020 and `ESM20` from around 3/18/2020 to around 6/18/2020.

Term Project: Volume-based Order Management

3 definitions & notations

The following subsections will cover notions that we will be using throughout the project.

3.1 holding period/time interval

At any time t, we assume a time horizon or holding time τ. For this study, τ is user specified. You may assume $5, 10, 15, 20, 25, 30, 45, 60$ min holding times. x_t at time t of the interval $[t, t+\tau)$ is the open price which is independent of the holding period. For example for S&P 500 Mini futures at time $t = 09:50:00$, $x_t = 3881.75$ that is the open price at that time irrespective of the holding time.

For this project, you many assume $\tau = 5, 10, 15, 30, 60$ min. Your module must be generic such that one can easily switch from one choice to another.

3.2 tick size

All futures contracts have a minimum price fluctuation known as a tick size also known as spread, ϵ. For example the tick size or spread for S&P 500 Mini futures is $\epsilon = 0.25$. For other futures markets you should be able to easily find the tick size (spread).

3.3 range

Range, $R_{t,\tau}$, is defined to be difference between high, H, and low, L for the interval $[t, t+\tau)$

$$\begin{aligned} R_{t,\tau} &= H - L \\ &= \ell\epsilon \end{aligned}$$

where ℓ is the number of spreads in the interval $[t, t+\tau)$.

The range for the previous interval will be written as $R_{t-\delta,\tau}$ which would be the range for the interval $[t-\delta, t-\delta+\tau)$ for some δ. In your analysis if t turns out to be the time that you are forecasting, have to make sure the last range would be non-overlapping otherwise you are forward looking that is $[t-\tau, t)$ should be the last one in your analysis which is not overlapping with $[t, t+\tau)$.

For example for time $t = 09:50:00$ and holding period $\tau = 1$ minute, the high and low price are $H = 3882.25$ and $L = 3881.25$ respectively, therefore

$$\begin{aligned} R_{09:50:00,1-min} &= H - L \\ &= 3882.25 - 3881.25 \\ &= 4\epsilon \end{aligned}$$

On the other hand for the same time but holding time of $\tau = 5$ min, the high price is $H = 3882.75$ and the low price is $L = 3879.25$ and we get

$$\begin{aligned} R_{09:50:00,5-min} &= H - L \\ &= 3882.75 - 3879.25 \\ &= 14\epsilon \end{aligned}$$

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3.4 rangeUp & rangeDn

RangeUp, $R^{(U)}_{t,\tau}$, is defined to be difference between high, H, and open, O for the interval $[t, t+\tau)$

$$\begin{aligned} R^{(U)}_{t,\tau} &= H - O \\ &= \ell_1 \epsilon \end{aligned}$$

where ℓ_1 is the number of spreads between H and O, in the interval $[t, t+\tau)$.

RangeDn, $R^{(D)}_{t,\tau}$, is defined to be difference between open, O, and low, L for the interval $[t, t+\tau)$

$$\begin{aligned} R^{(D)}_{t,\tau} &= O - L \\ &= \ell_2 \epsilon \end{aligned}$$

where ℓ_2 is the number of spreads between O and L, in the interval $[t, t+\tau)$.

Easy to see the range can be written as the sum of rangeUp & rangeDn

$$\begin{aligned} R_{t,\tau} &= H - L \\ &= H - O + O - L \\ &= R^{(U)}_{\tau} + R^{(D)}_{\tau} \end{aligned}$$

For example for time $t = 09{:}50{:}00$ and holding period $\tau = 5$ minute, the high price of $H = 3882.75$, the low price of $L = 3879.25$, open price of 3881.75, we will have

$$\begin{aligned} R^{(U)}_{09:50:00,5-min} &= H - O \\ &= 3882.75 - 3881.75 \\ &= 4\epsilon \end{aligned}$$

and

$$\begin{aligned} R^{(D)}_{09:50:00,5-min} &= O - L \\ &= 3881.75 - 3879.25 \\ &= 10\epsilon \end{aligned}$$

4 `ePDF` of range, rangeUp, and rangeDn for given τ

At any time t and time horizon τ we are interested in finding the following probability based on historical `OHLC` for interval $[t_i, t_i + \tau)$ for all t_i prior to t that is

$$P(R_{t,\tau} = \ell\epsilon) \quad \text{for} \quad \ell = 0, 1, 2, \ldots$$

as well as

$$\begin{aligned} P(R^{(U)}_{t,\tau} = \ell_1\epsilon) &\quad \text{for} \quad \ell_1 = 0, 1, 2, \ldots \\ P(R^{(D)}_{t,\tau} = \ell_2\epsilon) &\quad \text{for} \quad \ell_2 = 0, 1, 2, \ldots \end{aligned}$$

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5 ePDF of range, rangeUp, and rangeDn for given τ conditioning on the state

Quantities $R_{t,\tau}, R^{(U)}_{t,\tau}$, and $R^{(D)}_{t,\tau}$ not only depend on volume and volatility, it may also depend on the previous state we were in, locally up-trend or down-trend.

To be able to achieve that, we construct recursive running volatility and volume which could be based on simple moving or exponential weighted moving averages and binning them into different regimes/states on the run. A recursive formula for exponential weighted moving mean and variance is shown in Algorithm 1.

Algorithm 1: Exponentially weighted moving mean & variance (EWMA & EWMV)

```
set m
set half-life λ = 2^(-1/m)
for j = 1, ..., n do
    if j == 1 then
        sumW_j = 0
        sumWX_j = 0
        ewma_j = 0
        sumWSS_j = 0
        ewmv_j = 0
    else if j == 2 then
        sumW_j = 1
        sumWX_j = η_{j-1}
        ewma_j = sumWX_j / sumW_j
        sumWSS_j = (η_{j-1} - ewma_j)^2
        ewmv_j = sqrt(sumWSS_j / sumW_j)
    else
        sumW_j = λ × sumW_{j-1} + 1
        sumWX_j = λ × sumWX_{j-1} + η_{j-1}
        ewma_j = sumWX_j / sumW_j
        sumWSS_j = λ × sumWSS_{j-1} + (η_{j-1} - ewma_j)^2
        ewmv_j = sqrt(sumWSS_j / sumW_j)
    end
end
```

Note in Algorithm 1 at any step j we use η_{j-1} to avoid forward looking calculation otherwise our analysis would not be fully out-sample. η_j can be range, rangeUp, rangeDn, volume and any other quantity you may consider for the time interval $[t_j, t_j + \tau)$.

Having constructed these quantities recursively, we can calculate the following probabilities and keep updating them as new data arrives

$$P(R_{t_j,\tau} = \ell\epsilon \mid v_{j-1} \in m, \sigma_{j-1} \in n, \Delta x_{j-1} \in k) \quad \text{for} \quad \ell = 0, 1, 2, \ldots \& j = J_s, \ldots, J_e$$
$$P(R^{(U)}_{t_j,\tau} = \ell\epsilon \mid v_{j-1} \in m, \sigma_{j-1} \in n, \Delta x_{j-1} \in k) \quad \text{for} \quad \ell = 0, 1, 2, \ldots \& j = J_s, \ldots, J_e$$
$$P(R^{(D)}_{t_j,\tau} = \ell\epsilon \mid v_{j-1} \in m, \sigma_{j-1} \in n, \Delta x_{j-1} \in k) \quad \text{for} \quad \ell = 0, 1, 2, \ldots \& j = J_s, \ldots, J_e$$

where v_{j-1} is the volume in the interval $[t_{j-1}, t_{j-1}+\tau) = [t_j-\tau, t_j)$ being in state m for $m = 1, \ldots, M$, σ_{j-1} volatility at time t_{j-1} being in state m for $m = 1, \ldots, M$, and $\Delta x_{j-1} = x_j - x_{j-1}$ being at state k for $k = 1, \ldots, K$. J_s is defined by user and it should be large enough to start constructing the empirical probability density functions and J_e is dictated by length of data (end of data). Number of states/regimes N, M, K are defined by the user as well and depends on holding time, τ, as well.

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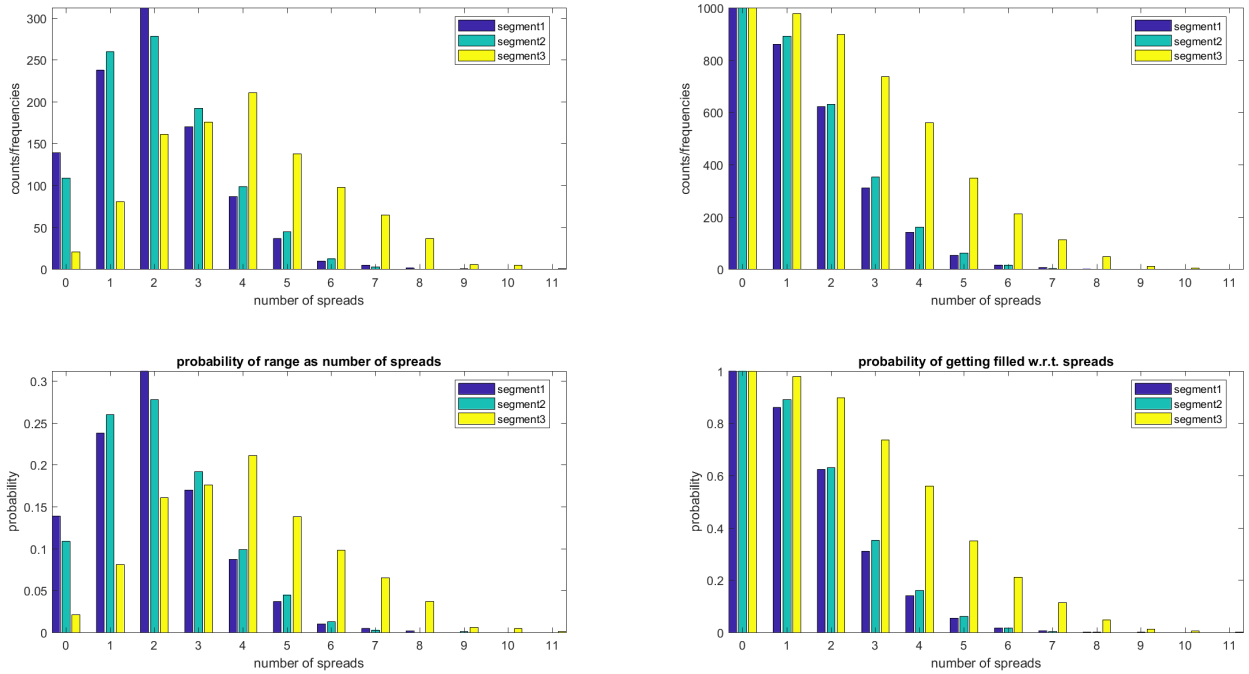

Figure 2: top-left plot: counts/frequencies as a function of number of spreads; bottom-left plot: empirical probability distribution function of rangeDn as a function of number of spreads; top-right plot: counts/frequencies of getting filled with respect to number of spreads; bottom-right plot: probability of getting filled with respect to number of spreads

In Figure 2, we illustrate an empirical probability distribution function of a futures contract for three different segments of data without any classification i.e. no conditioning on any state. Note that the empirical pdf for segments 1 & 2 are very similar and very different from segment 3. In Figure 2 we also illustrate probability of a limit order getting filled at a certain level as a function of spread. In order to apply Bayesian inference, we just use probability of market going up/down by certain number of spreads as a function of spread.

Term Project 2 Volatility-Volume-t

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